Non-Linear Dynamics Notes

Vyas Mokal

January 12, 2026

Contents

Lecture 1

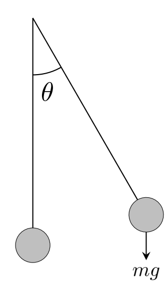
1.1 Books

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed., Westview Press, 2015.

This is Bible for the subject. It is standard, detailed and a bit old text.

1.2 Dynamics

- Study of evolution of systems
  Eg. Simple Pendulum

  has equation of motion
  $$\ddot{\theta} - \frac{g}{l}\sin\theta = 0 \tag{1.1}$$
  For small anlges
  $$\ddot{\theta} - \frac{g}{l}\theta = 0 \tag{1.2}$$
  Which is linear. But if θ is not small then non-linear and hence included in NLD[1].

- As equation becomes Non-Linear, the solution drastically changes

[1] **NLD:** Non-Linear Dyanmics

- Non-Negligible Non-Linear terms are not perturbative and needs separate study
- The subject NLD has advanced after the computers became powerful enough to perform high level computation

1.3 1D Maps

- Suppose we have a sequence of numbers (real numbers mostly)

$$x_0, x_1, x_2, \cdots$$

Such that

$$x_0, x_1 = f(x_0), x_2 = f(x_1), \cdots \tag{1.3}$$

i.e.

$$x_{n+1} = f(x_n) \tag{1.4}$$

Then it is called an *Map*[2]

1.3.1 Fixed Point

- Solution of (1.3) such that

$$x^* = f(x^*) \tag{1.5}$$

then x^*[3] is called *fixed point*. Note that it will give rise to the sequence of form

$$x_0, x_1, x_2, \cdots, x^*, x^*, x^*, \cdots$$

1.3.2 Statbility of a Fixed Point

- Suppose $x_0 = \bar{x} + \delta$.

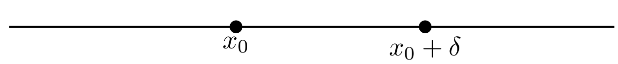


Fig. 1

- Now if the subsequent x's approach the fixed point x^* then the fixed point is stable and called an *attractor*

[2] Note here that the function f remains same
[3] It is widely followed conventions to label fixed point as x^* and we'll continue using it

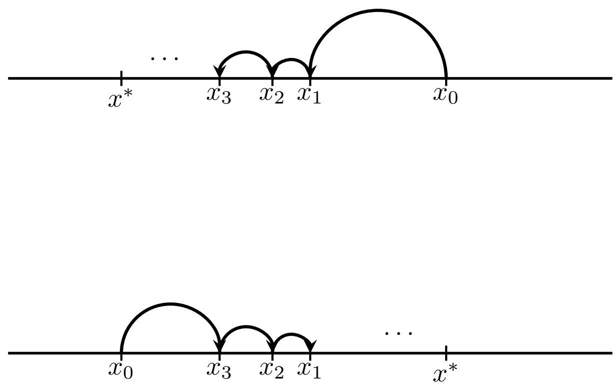


Fig. 2

- Approach to x^* is Oscillatory and is still fixed point

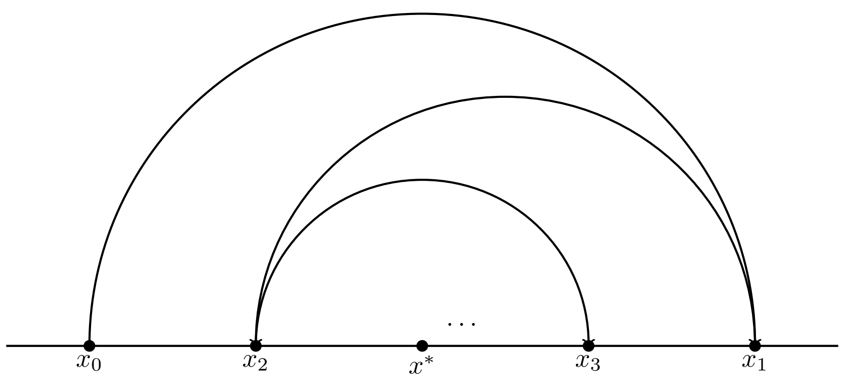


Fig. 3

- If the subsequent points diverge from the fixed point (monotonically or oscillatory) then the point x^* is Unstable Fixed Point

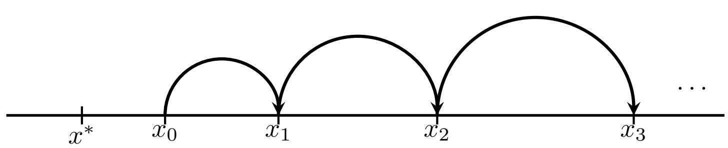


Fig. 4

1.3.3 Digramatic way to find number of fixed points

Follow the steps:

1. Draw the line $y = x$

2. Draw the curve $y = f(x)$

3. No. of intersections = No. of Fixed Points[4]

Example 1. $\mathbf{f}(\mathbf{x}) = \sin \mathbf{x}$

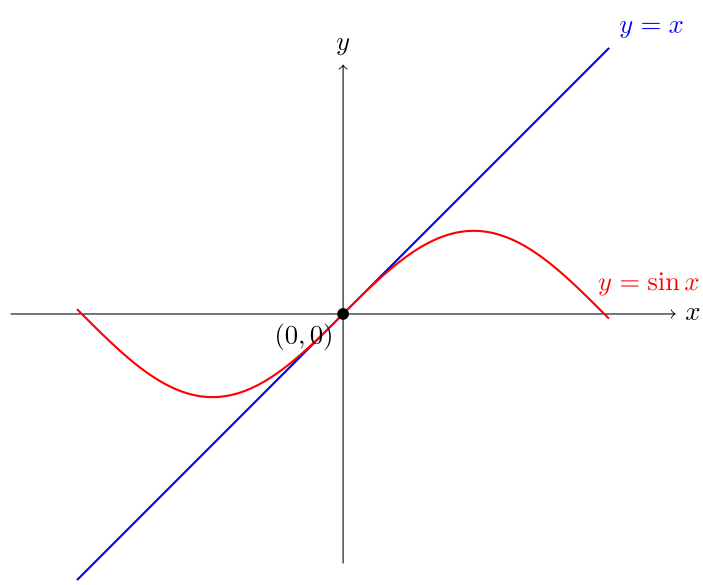


Fig. 5

There is only one fixed point at $x = 0$

Example 2. $\mathbf{f}(\mathbf{x}) = \cos \mathbf{x}$

[4]This logic is clear from the difinition of the fixed point as (1.3.1)

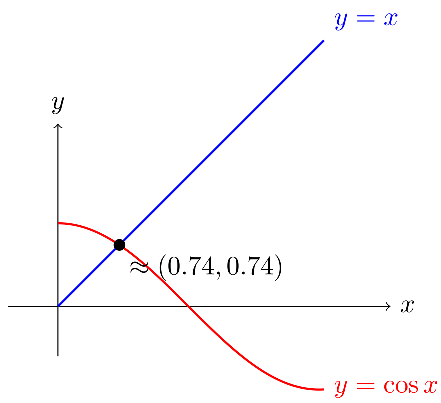


Fig. 6

There is only one fixed point at $x \approx 0.74$

Example 3. $\mathbf{f(x) = \frac{2x}{1+x}}$

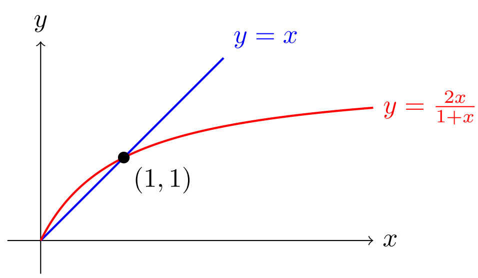


Fig. 7

There is only one fixed point at $x = 1$

- In monotonic approach
  $|x_{n+1} - x^*| < |x_n - x^*| \implies$ Stable fixed point
  $|x_{n+1} - x^*| > |x_n - x^*| \implies$ Untable fixed point

- We define the quantity *multiplier* by

$$\mu = \frac{x_1 - x^*}{x_0 - x^*} \tag{1.6}$$

It determines whether the fixed point is stable or unstable.

$$|\mu| < 1 \implies \text{Stable} \tag{1.7}$$

$$|\mu| > 1 \implies \text{Unstable} \tag{1.8}$$

$$\mu > 0 \implies \text{Monotonic} \tag{1.9}$$

$$\mu < 0 \implies \text{Oscillatory} \tag{1.10}$$

- Now, let's analyze the fixed point analytically. Let

  $$x_0 = x^* + \delta \tag{1.11}$$

  Then

  $$\begin{aligned} x_1 = f(x_0) &= f(x^* + \delta) \\ &= f(x^*) + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \cdots \\ &= x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \cdots \end{aligned}$$

- Now linearize

  $$x_1 = x^* + \delta f'(x^*) \tag{1.12}$$

  Therefore,

  $$f'(x^*) = \frac{x_1 - x^*}{\delta} = \frac{x_1 - x^*}{x_0 - x^*}$$

  Using (1.6)

  $$\mu = f'(x^*) \tag{1.13}$$

- Now in Example 1: $\mu = \cos x^* = 1$. The case $\mu = 1$[5] needs special analysis and we skip it for a moment

- In Example 2: $\mu = -\sin(x^*) < 0$, hence fixed point is stable and oscillatory.

- In Example 3:

  $$\mu = \frac{2(1+x^*) - 2x^*}{(1+x^*)^2} = \frac{2}{(1+x^*)^2}$$

  At $x^* = 0$; $\mu = 2 \implies$ monotonically unstable
  At $x^* = 1$; $\mu = 1/2 \implies$ monotonically stable

- Calculating Stability form multiplier is only valid when Linearization is valid.

- We now adopt Diagramatic Approach: *Cobwebb Diagrams*

  1. Draw curve $y = x$
  2. Draw $y = f(x)$

[5] In this case the fixed point x^* is called *Marginally Stable*

3. Draw line from x_0 to $f(x)$
4. Draw horizontal line to $y = x$ and that point of intersection gives x_1

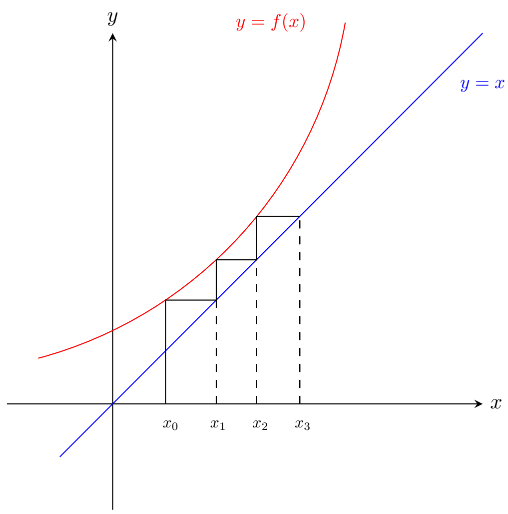


Fig. 8

The figure above depicts an example.

Now let's draw Cobwebb diagrams to our three examples.

- For Example 1

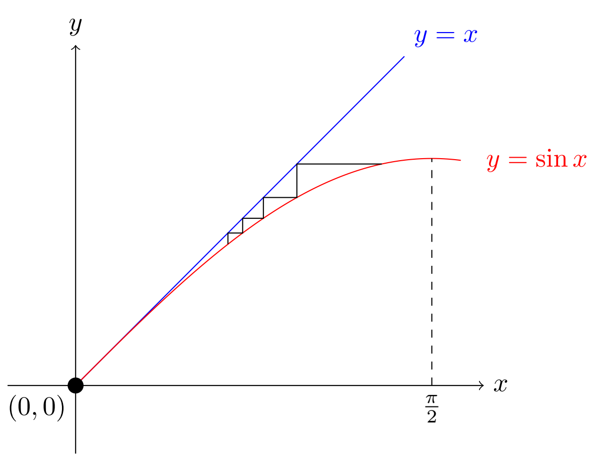


Fig. 9

Note that Cobwebb diagram shows $x = 0$ as stable fixed point unlike out analytical approach which failed due to $\mu = 1$.

- For Example 2

$\mu = 0$ **and** $\mu = 1$ **Case:**

$$x_1 = x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \cdots$$

- for $\mu = 1$

$$\begin{aligned} x_1 &= x^* + \delta + \frac{1}{2}\delta^2 f''(x^*) + \cdots \\ &= x_0 + \frac{1}{2}\delta^2 f''(x^*) + \cdots \end{aligned} \tag{1.14}$$

hence from above the higher order terms are important and will determine stability.

- For $\mu = 0$

$$\begin{aligned} x_1 &= x^* + \frac{1}{2}\delta^2 f''(x^*) + \cdots \\ &\approx x^* + c\delta^2 \qquad \left\{c = \frac{1}{2}f''(x^*)\right\} \end{aligned} \tag{1.15}$$

Now,

$$x_2 = x^* + c(c\delta^2)$$

And hence,

$$x_n = x^* + c^{(\cdots)}\delta^{2^n} \tag{1.16}$$

Since convergence is very fast, it is *superstable* fixed point.

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$$\begin{aligned} x_1 &= f(x_0) = f(x^* + \delta) \\ &= x^* + \delta f'(x^*) + \frac{1}{2}\delta^2 f''(x^*) + \cdots \end{aligned}$$

Linearize

$$\begin{aligned} x_1 &= x^* + \delta\mu \\ x_2 &= f(x_1) = f(x^* + \delta\mu) \\ &= x^* + \mu^2\delta \\ x_3 &= \mu^3\delta \end{aligned}$$

$$x_n = x^* + \mu^3\delta \tag{1.17}$$

$$\begin{aligned} \frac{x_n - x^*}{x_0 - x^*} &= \mu^n = e^{n\log\mu} \qquad (\mu > 0) \\ &= (|\mu|\text{sign}(\mu))^n \\ &= [\text{sign}(\mu)]^n e^{n\log|\mu|} \end{aligned}$$

$$x_n = x^* + \delta.\,(\text{sign}(\mu))^n\, e^{n\log|\mu|} \tag{1.18}$$

1.4 Logistic Map

We define *Logistic Map*[6][7] by

$$f(x) = rx(1-x) \qquad r > 0 \tag{1.19}$$

- Now let's find the fixed points

$$x^* = rx^*(1-x^*)$$

$$\boxed{x^* = 0} \tag{1.20}$$

$$1 = r(1-x^*)$$

$$\boxed{x^* = 1 - \frac{1}{r}} \tag{1.21}$$

- We'll always keep $0 \leq |x_n| \leq 1$ and vary $r > 0$
  $x^* = 0$ exist for all r
  $x^* = 1 - \frac{1}{r}$ exist for $r \geq 0$

- Stability of Fixed Points

$$f'(x) = r - 2rx$$

  At $x^* = 0 \quad f(x^*) = r \implies$ stable fixed point for $r < 1$ and unstable for $r > 1$; $r = 1$ is marginally stable
  At $x^* = 1 - \frac{1}{r}$

$$\begin{aligned} f(x^*) &= r - 2r\left(1 - \frac{1}{r}\right) \\ &= r - 2r + 1 \\ &= 2 - r \end{aligned}$$

  for $r = 1 \quad \mu = -1$ marginally stable
  for $r = 3 \quad \mu = 1$ marginally stable

- for $r > 3$ (say $r = 3.1$) we get sequence

$$x_0, x_1, \cdots, p, q, p, q, \cdots \tag{1.22}$$

  This alternate sequence of $p's$ and $q's$ is called a *2- cycle*[8].

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[6] $f(x)$ is quadratic hence included in NLD
[7] This equations appears in many areas: Engineernig, Population Dynamics, etc.
[8] A fixed point is a 1-cycle

Lecture 2

2.1 Logistic Map (cont...)

- For stability

$$\begin{aligned} -1 < \mu < 1 &\implies -1 < 2 - r < 1 \\ &\implies -3 < -r < -1 \\ &\implies 1 < r < 3 \end{aligned}$$

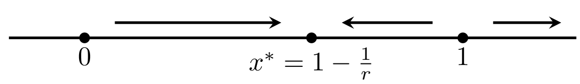

Fig. 2.1

- for a 2-cycle

$$f(p) = q \qquad f(q) = p \tag{2.1}$$

$$\therefore f(f(x)) = f(x) = x$$

and solution must be p and q

$$\begin{aligned} r(rx(1-x))(1 - rx(1-x)) = x \\ r^2 x(1-x)(1 - rx(1-x)) = x \end{aligned}$$

$f(x^*) = x^*$ and therefore $f(f(x^*)) = f(x^*) = x^*$
Therefore, $0, 1 - \frac{1}{r}$ are solution of above equation

$$x\left[r^2(1-x)(1 - rx(1-x) - 1)\right] = 0$$

taking out $x = 0$ factor

$$r^2(1-x)(1 - rx(1-x) - 1) = 0$$

put $r(1-x) = y \implies rx = r - y$

$$\begin{aligned} ry(1-xy) - 1 &= 0 \\ yr - y^2 rx - 1 &= 0 \\ yr - y^2(r-y) - 1 &= 0 \\ (y-1)(-yr + y^2 + y + 1) &= 0 \end{aligned}$$

$$y = 1 \implies r(1-x) = 1 \implies x = 1 - \frac{1}{r}$$

$$y^2 + y(1-r) + 1 = 0 \tag{2.2}$$

Now, consider

$$y = \frac{-(1-r) \pm \sqrt{(1-r)^2 - 4}}{2} \tag{2.3}$$

$$= \frac{(r-1) \pm \sqrt{r^2 - 2r - 3}}{2} \tag{2.4}$$

$$= \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \tag{2.5}$$

$$r(1-x) = \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2} \tag{2.6}$$

$$x = 1 - \frac{(r-1) \pm \sqrt{(r-3)(r+1)}}{2r} \tag{2.7}$$

$$= \frac{1}{2r}\left(r + 1 \pm \sqrt{(r-3)(r+1)}\right) \tag{2.8}$$

$$\tag{2.9}$$